



Risks and Risk Management in Infrastructure Projects

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Agenda



- Risks in Infrastructure and Risk Management
- Quantitative Risk Analysis
- Qualitative Risk Management
- Risk Management Strategies

Risk..



- Possibility of something bad happening
- Risk management is increasingly a critical success factor for major infrastructure projects.
- Risk management is often governed by the principle that it should be borne by the party most capable of controlling it.
- Risk is different from uncertainty. In case of uncertainty there exists more than one possible outcomes but the probability for a particular outcome to occur is not known.
- A risk left unidentified is difficult to mitigate in later stages of the projects so the concept of risk management starts with identifying the potential risks.

Risk profile of various infrastructure sectors



- **Power Sector** –illegal connection, demand risk, technical and environmental risk, variation in cost and availability of fuel
- **Road Transport**–fluctuation of actual traffic from forecasted traffic, cost overruns and delays
- **Tunnel and Bridge**– Safety, Health, Weather
- **Rail Transport** – expensive rolling stock and control system
- **Airports and Ports**– Trade, Connectivity to other modes of transport, political risk

Classification of Risk..



- Completion risks
- Performance risks — poor design or inadequate technology
- Operation & maintenance risks
- Financing risks
- Revenue risks
- Input supply risks

Risk Management



- Risk identification: The process of identifying all the risks associated with the project, whether during its development phase, or its construction or operational phase.
- Risk assessment: The process of determining the likelihood of the identified risks materialising and the magnitude of their consequences if they do materialise.
- Risk allocation: The process of allocating responsibility for dealing with the consequences of each risk to one of the project stakeholders, or agreeing to share the risks.
- Risk mitigation: The process of attempting to reduce the likelihood of the risk occurring and the degree to its consequences for the risk-taker

Snapshot of Financial loss due to different Risks



Example	Budget vs actual € billion		Delays and start-up problems	Incorrect capacity and revenue plans	Total value lost vs plan € billion
	Planned	Actual			
Eurotunnel	7.5	15.0	6-month delay 18 months of unreliable service after opening	Overestimated market-share gain in freight and passengers by 200%	~7.5
High-speed rail Frankfurt-Cologne	4.5	6.0	1-year delay of construction - Legal and technical issues	Unforeseen capped government funding	~1.5
Betuwe Line NL (cargo rail)	2.3	>5.0	1.5-year¹ delay of construction - Technology choices still not finalized	Annual revenue shortfall of €20 million	~3.0
Kuala Lumpur Airport	2.0	3.5	- Initial issues with connectivity to downtown area - Complaints about facility hygiene levels	- Handles only ~60% of current capacity - Losing market share to Singapore	~1.5

¹ Project still not finalized and costs could go even higher.

Source: Annual reports; Reuters; Jane's Airport Review; McKinsey analysis

Major Risks in Infrastructure Projects



- Risk of Expropriation (Project Development)
- Project Completion Risk (Construction Period)
- Technology Risk (Construction/Operation Period)
- Operations and Maintenance risk (Operation Period)
- Interest Rate Risk (Operation Period)
- Inflation Risk (Operation Period)
- Force Majeure (Throughout Project Period)
- Social risks (Throughout Project Period)

Major categories of risks in Infrastructure Projects



<u>Political Risks E.g.</u> Risk of Expropriation Contract Reneging Cancellations Delays in Permits and Approvals	<u>Economic Risks E.g.</u> Currency Devaluations Inflation and lowering of purchasing power Demand Forecast Errors
<u>Socio-Economic Risks E.g.</u> Community Protests Unwillingness to Pay	<u>Technological Risks E.g.</u> Construction Delays Inefficiencies in Operations and Maintenance



RISK UNIVERSE

Major risk categories in infrastructure projects



Planning & management

Poor scope definition, weak scheduling, interface gaps, and unrealistic baselines.



Technical & operational

Design errors, constructability issues, technology failure, and poor quality control.



Financial & legal

Funding gaps, insolvency, claims, contract disputes, inflation, and currency pressure.



Human & stakeholder

Labor shortages, safety issues, resistance from communities, and weak stakeholder alignment.



Environmental & external

Weather shocks, land acquisition delays, permits, utilities, and force majeure events.



Strategic & political

Policy changes, regulatory shifts, elections, geopolitical friction, and reputational risk.

Tools and Techniques for Risk Identification



- Checklist of Risk
- Review of Project Documentation
- Detail Identification of Risks by experienced technical person-
 - Brainstorming
 - Delphi Technique
 - Interviews
 - SWOT
 - Diagramming- Cause and Effect diagram

Output of Risk Identification



- List of Identified Risks
- List of potential responses
- Root cause of Risk
- Updated Risk Categories
- Risk Register

Risk Assessment



- Qualitative Risk Analysis - linguistic
- Quantitative Risk Analysis

Illustrative risk matrix



	Low impact	Moderate impact	High impact	Severe impact
Low probability	Minor permit delay	Small rework item	Localized utility clash	Late approval on key package
Medium probability	Temporary labor shortage	Design coordination gap	Commodity cost escalation	Land handover delay
High probability	Weather disruption	Scope change	Interface failure	Funding disruption / claim spiral

The objective is to separate non-critical risks from risks that require immediate mitigation, contingency planning, and senior management oversight.

Typical high-priority risks and relative attention





RESPONSES

Five basic treatment strategies



Avoid

Change scope, alignment, technology, or delivery method to eliminate exposure.



Reduce

Strengthen design, staging, supervision, quality assurance, and early stakeholder engagement.



Transfer

Use insurance, guarantees, hedging, and contract clauses to shift exposure.



Share

Allocate risk through PPP structures, alliances, and collaborative commercial mechanisms.



Accept

Retain low-priority risk with reserves, contingencies, and predefined trigger actions.



Review continuously

Responses must be updated as the project moves from design to construction and operations.

Impact Risk Rating Matrix



Impact	Probability				
	Very High	High	Medium	Low	Very Low
Catastrophic	High	High	Moderate	Moderate	Low
Critical	High	High	Moderate	Low	None
Marginal	Moderate	Moderate	Low	None	None
Negligible	Moderate	Low	Low	None	None

Probability/Impact Risk Matrix



Probability	Probabilty (100)		
	Low (10)	Medium (50)	High
High (1.0)	Low Risk $10 \times 1.0 = 10$	Medium Risk $50 \times 1.0 = 50$	High Risk $100 \times 1.0 = 100$
Medium (0.5)	Low Risk $10 \times 1.0 = 10$	Medium Risk $50 \times 0.5 = 25$	Medium Risk $100 \times 0.5 = 50$
Low (0.1)	Low Risk $10 \times 1.0 = 10$	Low Risk $50 \times 0.1 = 5$	Low Risk $100 \times 0.1 = 10$

Decisioneering Approach



- You are building a toll road
- H,M,L correspond to three different traffic volumes conditional on the state of the economy (According to your Expert)

State of the Economy	Probability
Good	0.25
OK	0.45
Bad	0.3

Pay-offs



- Your Initial Investment
= \$1,000,000

Traffic Volumes	Revenue
H(igh)	\$1,400,000
M(edium)	\$1,200,000
L(ow)	\$500,000

Expert's Conditional Probabilities

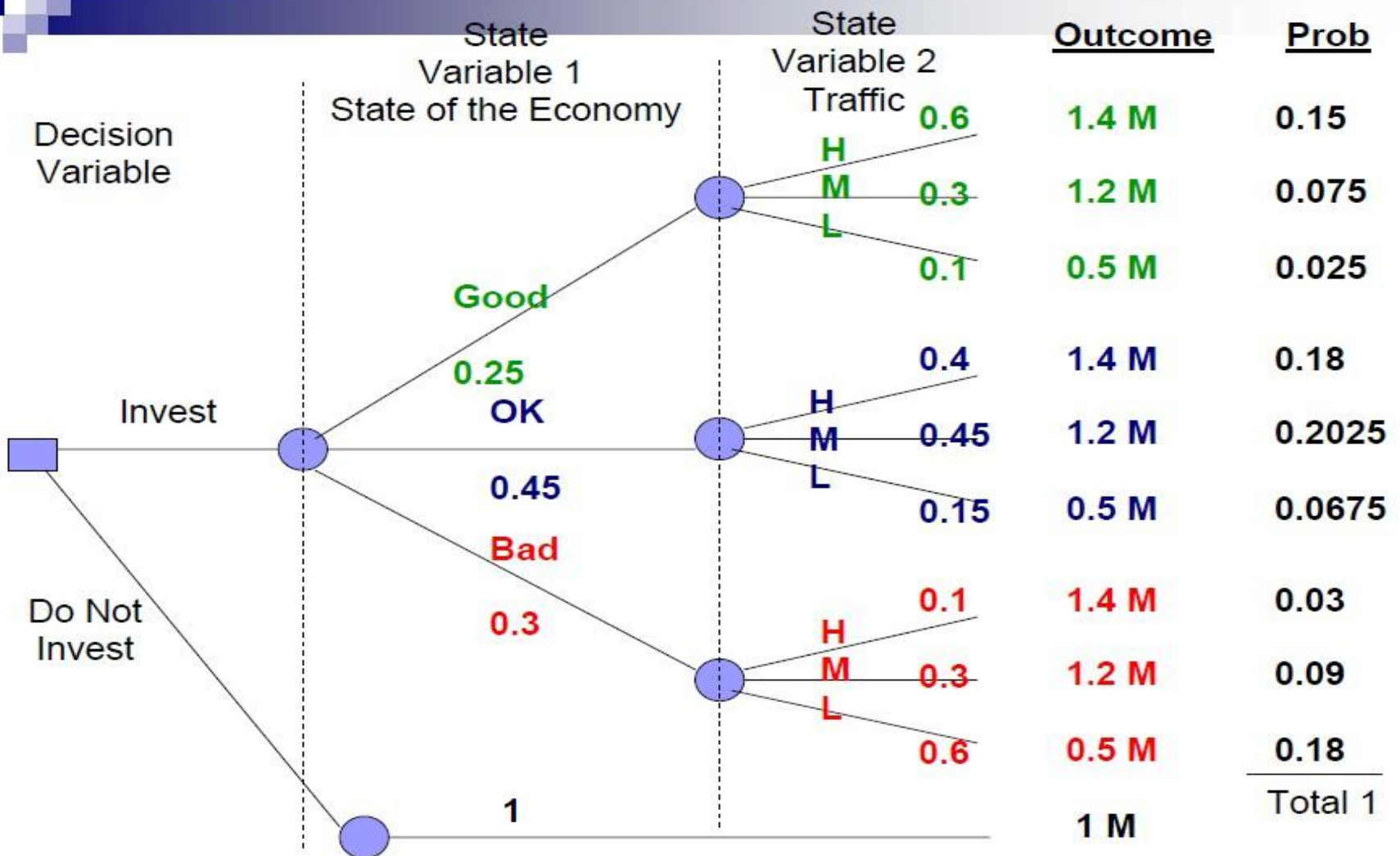
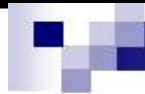


Economy vs Traffic	Good	OK	Bad
High	0.6	0.4	0.1
Medium	0.3	0.45	0.3
Low	0.1	0.15	0.6



Will You Invest?

Why or Why Not?



Expected Value of the Investment



Expected Value of the Investment

- $EV = \Sigma(\text{Outcome} * \text{Probability})$

- $= 1.4 (0.15 + 0.18 + 0.03) + 1.2 (0.075 + 0.2025 + 0.09) + 0.5(0.025 + 0.0675 + 0.18)$

$$= \$1,081,250$$

Since this value is greater than the initial Investment of \$1,000,000 a risk neutral investor will invest

Sensitivity Analysis for Adverse Changes



Independent Variable	Base Case Scenario	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Worst Case Scenario
Capital Expenditure increase		15%				15%
Operating Expenditure increase			10%			10%
Traffic volume decrease				15%		15%
Interest Rate Increase					5%	5%
Dependent Variable						
IRR	17%	14%	15%	13%	16%	12%

Sensitivity Analysis for Beneficial Changes



Independent Variable	Base Case Scenario	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Best Case Scenario
Capital Expenditure decrease		5%				5%
Operating Expenditure decrease			9%			9%
Traffic volume increase				12%		12%
Interest Rate decrease					2%	2%
Dependent Variable						
IRR	17%	18%	17.5%	19%	17.2%	19.8%

Risk Management



1. Accept Risk

The stakeholders who are responsible for a risk can choose to accept a risk. For example, the risk that a project may fail may be accepted if the project is of strategic importance.

2. Mitigate Risk

Actions are taken to reduce risk to an acceptable level. For example, the organization assigns a top performing project management team to a project to reduce the risk that it will fail.

The risk that remains after mitigation is known as residual risk. Residual risks are also controlled (accepted, mitigated, eliminated or transferred)

3. Eliminate Risk

A risk may be reduced to zero. Normally the only way to accomplish this is to cease the activity that generates the risk. For example, selling a risky investment will eliminate the risks associated with that investment.

4. Transfer Risk

A risk may be transferred to another organization or individual. For example, fire insurance transfers the risk of asset damage due to fire.

Risk Management



- The probabilistic approach is useful to estimate and quantify risks and make decisions on whether or not to go ahead.
- However to manage the more “fuzzy” political and socioeconomic risks, the “Managerial Model” depicted in the next two slides (taken from “The Strategic Management of Large Engineering Projects” by Miller and Lessard, can be more useful.

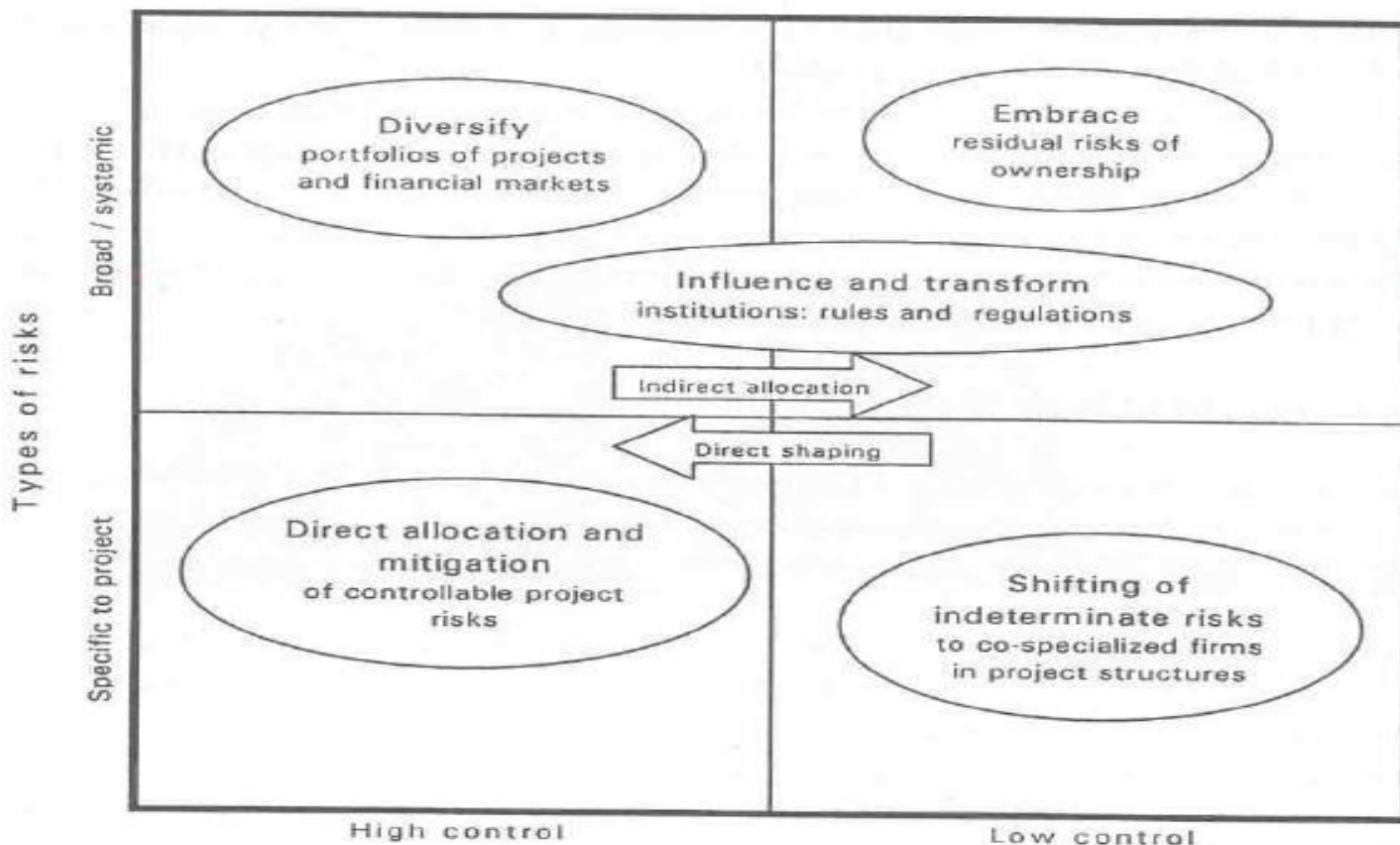
Risk Management



The Managerial Approach

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Lessard and Miller



The layering Model

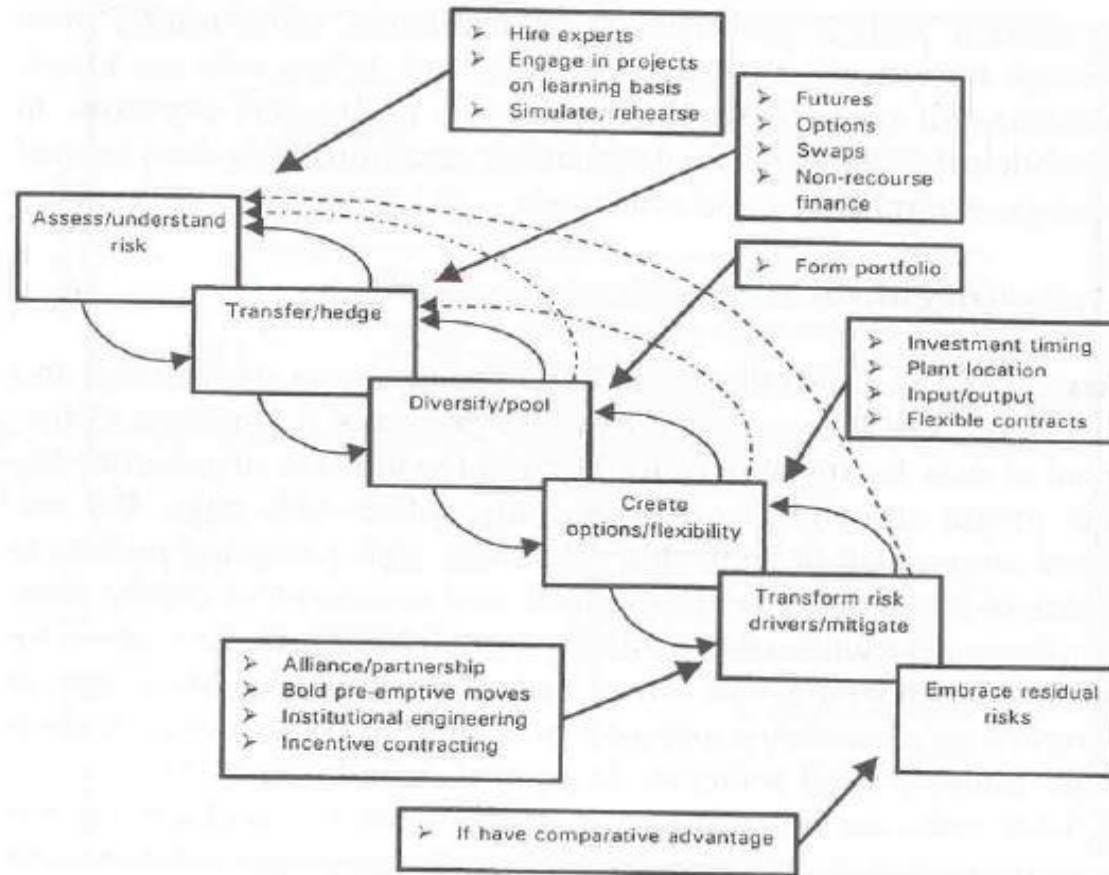


Figure 3.5
The layering model: strategizing to face risks

The Managerial Model explained



- In this model, all the various project risks are assessed and understood
- Risks that can be mitigated are mitigated. For instance, Construction Delays risks are mitigated by using a sophisticated schedule
- The residual risks are then first allocated to the party best able to bear those risks. E.g. Construction delay risk is allocated to the EPC contractor, demand risk may be allocated to the government if there are widespread economic fluctuations. This is usually done through the contract but can also be done relationally.

The Managerial Model explained



- If risks cannot be assigned to parties, the project organization can come up with a diversification strategy across many projects
- Risks not addressed by diversification can be addressed by lobbying and institutional transformation of the rules of the game.
- Any remaining risks will just have to be embraced.
- In this manner, an infrastructure project organization can effectively deal with risks that arise on an infrastructure project

Risk Management



RISK CATEGORY	PHASE OF PREDOMINANCE	RISK IDENTIFICATION	RISK MITIGATION
<i>Land Acquisition</i>	Project Development	Delays in land acquisition and in providing encumbered right of way to the EPC contractor can lead to delays in the start of construction resulting in escalation of project cost. Additionally, the risk of cost of acquisition not being contained within the estimates.	a) Timely provision of land for construction of facilities and systems should be made a condition precedent to the lease/concession Agreement. b) For delay related to consent, approvals, clearances, the government should grant necessary permissions and the same should be executed within specified time. c) On delay, the concession period should be extended by an equal period.
<i>Project Completion Risk</i>	Construction Period	The project completion risk or the contractor's risk refers to the possibility of non-completion of the project with the designated period from the Notice to Proceed. Any delays in the construction may be expected to result in increased construction costs.	This risk should be mitigated through a Provision under Concession Contract (CC) CC with EPC contractor to include turnkey, fixed price design & construction contract with payments made on reaching certain milestones. Contractor to pay Liquidated Damages for delays during construction. Independent Engineer should review and monitor progress.

Risk Management



<i>Technology Risk</i>	Construction/ Operation period	This pertains to the risk that the project may be either physically inappropriate to handle the projected demand or is inappropriately designed to meet local socio-economic needs and hence rectification of these design could escalate the O&M costs during the operations period	the project to be designed after a comprehensive analysis of the local conditions the construction should be carried out with strict penalties for noncompliance of the technical design by the contractor
<i>Operations and Maintenance risk</i>	Operation period	In the event of O&M costs exceeding the estimates used for the establishment of financial viability, residual cash flows for debt/equity services would be lower than anticipated there by affecting project returns.	Selection of O&M operator will be on the basis of competitive bidding the selection criteria should take into account of its past record, fiduciary responsibility exhibited in other assignment, financial strength etc. The O&M contract should provide for a fixed & a variable fee which could be based upon the O&M requirements set forth in the Concession Agreement. O&M Contract is a fixed price contract, with risk of cost over-runs to be borne by the O&M Contractor.

Risk Management



<i>Financial Risk</i>			
<i>Interest Rate Risk</i>	Operations Period	Determination of project viability is predicated on the existing interest rate scenario prevailing in the country. A drastic increase in the interest rate scenario may affect the debt servicing capability through project cash flows and significantly depress shareholder returns, even though the project may still achieve the designated return	The project should be financed on an optimal mix of fixed rate and floating rate instruments, to hedge the interest rate movement risk
<i>Inflation Risk</i>	Operation period	The tariff rates being inflation indexed, the projected revenues and consequently the achievement of the designated rate of return would be adversely affected in case the inflation rate is lower than what has been assumed in the financial model	Tariffs to be adjusted for inflation during operations as per formulae given in the concession agreement
<i>Force Majeure</i>	Throughout project cycle	This risk category deals with nonpolitical events of force majeure considered as "Acts of god" such as epidemics, natural disaster, earthquake, floods during construction phase affecting construction and such other events. The impact of these risks on construction and/or the project operations could range from minor to server, say in case of earthquake, where the damage may be server enough to render the facilities irreparable	Comprehensive insurance coverage In force majeure risks of types which are not insurable, the investors should get yield of certain percentage on equity on date of termination